



AWS KMS BOUNTY
OPEN TRACK

Hedera Flow

Tokenizing Trust for Electricity Billing

Verify your electricity before you pay.

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Industrial Web3 Infrastructure
DePIN • Hedera • Enterprise

The Electricity Billing Trust Crisis

Estimated Bills

Millions of users receive estimated bills they **cannot verify**. No transparency into how charges are calculated.

Blind Payment

Consumers are forced to **blindly pay** for electricity they didn't actually use. No recourse, no proof.

Utility-Controlled Data

Even with modern smart meters, data is controlled by **utilities—not users**. Zero transparency.

Core Insight

Today, electricity billing is a **black box**—you're told what to pay, not what you used.

- No verifiable usage records
- Disputes resolved in utility's favor
- Consumers bear the burden of proof

Tokenizing Trust

Hedera Flow turns usage into **tamper-proof, verifiable data on-chain**.



LEGACY

Postpaid

- 1 Snap a photo of your meter
- 2 OCR extracts the reading
- 3 Image hashed to IPFS for dispute resolution
- 4 Data anchored to Hedera



LIVE

Prepaid

- 1 Buy electricity units (kWh)
- 2 Seamless fiat-to-crypto dashboard
- 3 Balances tracked on-chain
- 4 Deductions recorded immutably



FUTURE

Smart Meters

- 1 Meter autonomously signs data
- 2 Direct blockchain transmission
- 3 Cryptographic proof of origin
- 4 No human manipulation possible



CORE INSIGHT

We don't replace meters—we make them trustworthy.

Enterprise Security, Real-World Scale

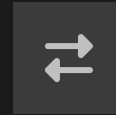


AWS KMS & Web2.5

Email login. AWS KMS generates Hedera wallets and signs transactions in the background.

No seed phrases

Enterprise-grade



EVM & HTS

Custom EVM logic for instant prepaid fiat-to-crypto payments (NGN to HBAR) via Hedera Token Service.

Instant settlement

Low fees

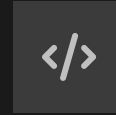


HCS & IPFS

Immutable log of every meter reading + IPFS for image verification and dispute resolution.

Tamper-proof

Decentralized storage



Pure gRPC Execution

Bypassed Java SDKs for pure grpcio Python execution. Lightweight and crash-proof architecture.

Lightweight

Crash-proof

AWS

KMS Integration

HCS

Consensus Service

HTS

Token Service

gRPC

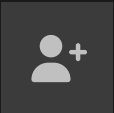
Native Execution

Driving Massive TPS & Account Growth



B2B2C Distribution

Partner with regional DisCos to slash manual meter-reading costs and end consumer disputes. Utilities benefit from reduced operational overhead.



Passive Onboarding

Email signups automatically spin up millions of AWS KMS-backed Hedera wallets. Zero friction for end users.



Network Success

Thousands of daily IoT pings (HCS) and automated payments (HTS/EVM) guarantee sustained, massive TPS for the Hedera network.

Network Growth Metrics

Daily IoT Pings (HCS) 10K+

Automated Payments (HTS) 5K+

Wallet Onboarding 1M+

TPS Contribution High

Projected Annual Impact

3.6M+

Transactions

Real Problem. Real Corporate Validation.

Consumer Validation

Hosted an X Space where real users confirmed the problem:

"We receive suspicious bills with no way to verify usage."

 Direct user feedback

Corporate Validation

Pitched the Hedera Flow architecture directly to:

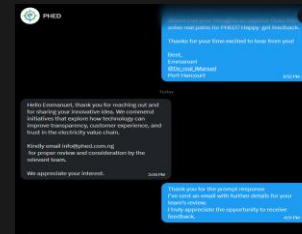
Port Harcourt Electricity Distribution Company

February 2026



The Result

They explicitly requested an official review of the system.



Official Response

"We commend initiative that explore how technology can improve transparency, customer experience, and trust in the electricity value chain."



This is not a hypothetical problem—it's a validated B2B solution.

The Global Utility Transparency Layer

01 Phase 1

Eliminate estimated electricity billing in **Nigeria**

 Primary Market

02 Phase 2

Cross-chain liquidity routing for seamless **local fiat payments**

 Multi-Chain

03 Phase 3

Expand to **Water, Gas, and Internet** billing globally

 Global Scale

“ AIM ”

No one should ever pay a bill they cannot verify.

Live Demo



hedera-flow-ivory.vercel.app



Demo Video

[Video Demo](#)

